Compression–Recovery Duality After Scrambling: Attractor-Class Reconstruction with Applications to Hopfield Networks, Teleportation, and Black Hole Information

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Abstract

We study a general reconstruction problem: a system's microstate is strongly scrambled (high entropy, locally thermal-looking), yet the system retains a latent basin/sector structure that permits recovery of *attractor class* from limited side information. We formalize this as a *compression–recovery duality*: when dynamics exhibit a family of attractor basins $\{\mathcal{A}_\mu\}_{\mu=1}^{M}$ in a state space $\mathcal{X}$, the information required to recover the correct basin label after scrambling scales as $O(\log M)$ rather than with the dimension of the microstate description. We prove a basin-exclusion lemma in a discrete Hopfield model, showing that clamping b coordinates defines a constraint manifold that excludes competing basins with probability $1 - 2^{-b}$, so $b \approx \log_2 M$ suffices to reduce ambiguity to $O(1)$. We then generalize the exclusion mechanism to continuous systems using *sector labels* $K : \mathcal{X} \to \mathcal{K}$ derived from global invariants (e.g., conserved charges, low-frequency spectral components), yielding an analogous bound in which $\log |\mathrm{range}(K)|$ replaces b. Finally, we explain how the black hole information problem can be cast in the same mathematical shape: microstates are scrambled by chaotic dynamics, local observables appear thermal, and recovery depends on global constraints/sector data rather than microstate-level access. The framework yields explicit scaling predictions and failure modes testable in toy models.

1 Introduction

A recurring technical pattern appears across several domains:

- A system undergoes *strong scrambling* (mixing, thermalization, decoherence, or chaotic dynamics), making microstate-level recovery appear impossible from local data.
- Despite scrambling, dynamics retain *structure*: attractor basins, superselection sectors, or constraint manifolds.
- A small amount of *side information*—a basin/sector label—is sufficient to guide the system into the correct class.

This motivates a reconstruction question that is deliberately *label-free*:

> Given a scrambled microstate, what minimal information is sufficient to recover the correct *attractor class*?

The key distinction is between: (i) *microstate specification*, which typically scales with system size, and (ii) *attractor-class specification*, which may scale only logarithmically (or even remain constant) if the number of relevant classes grows subexponentially.

We formalize this with a general theorem for attractor-based systems, then provide an explicit instantiation in a Hopfield network. We show that the exclusion mechanism has a clean probabilistic form (a constraint manifold removes most competing basins), and then generalize the same logic to continuous systems using global invariants (charges, spectra, coarse correlators) rather than pointwise clamping.

2 Problem Statement and Definitions

Definition 1 (State space and dynamics)**.** Let $\mathcal{X}$ be a state space (discrete or continuous). Let $\Phi_t : \mathcal{X} \to \mathcal{X}$ denote the system dynamics (e.g., gradient flow on an energy function, dissipative evolution, or a coarse-grained effective flow).

Definition 2 (Attractor basins)**.** A family of attractor basins is a collection $\{\mathcal{A}_\mu\}_{\mu=1}^{M}$ of subsets of $\mathcal{X}$ such that:

1. (basin invariance) if $x \in \mathcal{A}_\mu$, then $\Phi_t(x) \in \mathcal{A}_\mu$ for all $t \geq 0$;
2. (basin convergence) for x in a neighborhood of $\mathcal{A}_\mu$, $\Phi_t(x)$ converges to the attractor associated with $\mathcal{A}_\mu$.

We interpret $\mu \in \{1, \ldots, M\}$ as the *attractor label*.

Definition 3 (Scrambling map)**.** A scrambling map $\mathcal{S} : \mathcal{X} \to \mathcal{X}$ is a transformation that increases uncertainty about the attractor label when only local observables are available. Informally, $\mathcal{S}$ pushes typical states toward a high-entropy region of $\mathcal{X}$ and destroys easily accessible microstate structure.

Definition 4 (Side information and recovery)**.** Let $K : \mathcal{X} \to \mathcal{K}$ be a *label function* (side information), taking values in a finite or discretized set $\mathcal{K}$. Given a scrambled state $x' = \mathcal{S}(x)$, a recovery protocol uses $K(x)$ (or an estimate of it) to construct an initial condition $\tilde{x}_0$ and/or a biased evolution that converges to the correct basin $\mathcal{A}_{\mu(x)}$ with high probability.

Remark 1 (Information accounting)*.* If K takes values in a finite set $\mathcal{K}$, then the information content of the side information is

$$I(K) = \log_2 |\mathcal{K}| \quad \text{bits.}$$

For continuous K, we discretize to precision Δ; the effective information then scales as $\log_2 |\mathcal{K}_\Delta|$.

3 Compression–Recovery Theorem

We now state a general theorem capturing the scaling separation between microstate description and basin/sector identification.

Assumption 1 (Attractor identifiability by side information)**.** There exists a map $K : \mathcal{X} \to \mathcal{K}$ such that each basin $\mathcal{A}_\mu$ is concentrated in (or associated with) a small subset of labels. Formally, assume there exists a function $\kappa : \{1, \ldots, M\} \to \mathcal{K}$ such that for typical $x \in \mathcal{A}_\mu$, $K(x) = \kappa(\mu)$ with high probability.

Theorem 1 (Compression–recovery via basin labels)**.** *Assume $\{\mathcal{A}_\mu\}_{\mu=1}^M$ are attractor basins in $\mathcal{X}$, and there exists a side-information map $K : \mathcal{X} \to \mathcal{K}$ such that $\kappa(\mu) = K(x)$ for typical $x \in \mathcal{A}_\mu$ (i.e., K identifies basins up to degeneracy). Let $g := \max_{k\in\mathcal{K}} |\{\mu : \kappa(\mu) = k\}|$ be the maximum degeneracy (basins sharing the same label).*

Then:

1. *Any protocol that reduces the expected number of candidate basins to $O(1)$ requires at least*

$$I(K) \;\geq\; \log_2(M/g) \quad \text{bits}$$

   of side information (up to additive constants).

2. *If K is non-degenerate ($g = 1$) and can be extracted and imposed as a constraint/bias on reconstruction, then $O(\log_2 M)$ bits suffice to select the correct basin with $O(1)$ ambiguity, independent of microstate dimension.*

Remark 2 (Interpretation)*.* The theorem is intentionally agnostic about the physical realization of K. In discrete systems, K may be a coordinate clamp or a pattern label. In continuous systems, K may be a conserved charge, a topological sector, or a coarse spectral signature. The core claim is a scaling separation: basin ID scales like $\log M$, not like the microstate description length.

4 Discrete Instantiation: Hopfield Basin Recovery

4.1 Model

Consider a Hopfield network with N spins $s \in \{-1, +1\}^N$ and M stored patterns $\{\xi^\mu\}_{\mu=1}^M$. The weight matrix is

$$J_{ij} = \frac{1}{N}\sum_{\mu=1}^{M} \xi_i^\mu \xi_j^\mu, \qquad J_{ii} = 0,$$

and the energy is

$$E(s) = -\frac{1}{2}\sum_{i,j} J_{ij} s_i s_j.$$

Under standard asynchronous dynamics, each ξ^μ is an attractor (for M below capacity).

4.2 Lemma: Constraint-Manifold Basin Exclusion

Lemma 1 (Constraint-manifold basin exclusion)**.** *Let $\{\xi^\mu\}_{\mu=1}^M$ be patterns drawn i.i.d. uniformly from $\{-1, +1\}^N$. Fix a target $\mu^\star$ and a clamping set $I \subset \{1, \ldots, N\}$ of size $|I| = b$. Define the constraint manifold*

$$\mathcal{M}_I := \{s \in \{-1, +1\}^N : s_i = \xi_i^{\mu^\star} \text{ for all } i \in I\}.$$

Then for any competing pattern $\nu \neq \mu^\star$,

$$\mathbb{P}(\xi^\nu \in \mathcal{M}_I) = 2^{-b}.$$

Consequently, the expected number of competing patterns compatible with the constraint is

$$\mathbb{E}\Big[\,|\{\nu \neq \mu^\star : \xi^\nu \in \mathcal{M}_I\}|\,\Big] = (M-1)\,2^{-b}.$$

Proof. For each $i \in I$, $\mathbb{P}(\xi_i^\nu = \xi_i^{\mu^\star}) = 1/2$ by symmetry, independently across i. Thus $\mathbb{P}(\xi^\nu \in \mathcal{M}_I) = \prod_{i \in I}(1/2) = 2^{-b}$. Linearity of expectation yields the second claim. □

Corollary 1 (Logarithmic clamping suffices)**.** *To reduce the expected number of competing compatible attractors to $O(1)$ it suffices to choose*

$$b \approx \log_2 M.$$

In particular, if $M = \mathrm{poly}(N)$, then $b = O(\log N)$ constraints suffice.

4.3 Scramble–recover protocol (for experiments)

A concrete numerical protocol is:

1. Choose a target pattern $\mu^\star$ and initialize near $\xi^{\mu^\star}$.
2. Apply scrambling: replace s with a random spin configuration (or apply random flips at high temperature).
3. Impose side information either as (i) clamped coordinates on I or (ii) a global bias term that favors the target basin.
4. Run dynamics to convergence. Measure success by overlap with the target pattern.

Section A gives a full reference implementation and scaling plots.

5 Continuous Generalization: Sector Labels and Basin Exclusion

The Hopfield lemma clamps microscopic degrees of freedom. For continuous-field systems, a more physical analogue is to constrain *global invariants* or *low-dimensional signatures*.

Definition 5 (Sector label)**.** Let $K : \mathcal{X} \to \mathcal{K}$ be a sector label derived from invariants or coarse summaries of the configuration, for example:

$$K(x) = \big(Q(x),\, c_1(x), \ldots, c_K(x)\big),$$

where Q is a discrete conserved charge (or topological invariant) and c_k are low-frequency spectral coefficients of a memory field, correlation function, or other coarse observable.

Lemma 2 (Sector-constraint basin exclusion (uniform label model))**.** *Let $\{\mathcal{A}_\mu\}_{\mu=1}^M$ be attractor basins in $\mathcal{X}$. Suppose each basin has an associated label $\kappa(\mu) \in \mathcal{K}$ and basin labels are approximately uniformly distributed over $\mathcal{K}$ (up to degeneracy). Let x be a pre-scramble state in basin $\mathcal{A}_{\mu^\star}$ with label $K^\star := \kappa(\mu^\star)$, and define the sector-constraint set*

$$\mathcal{C}(K^\star) := \{x \in \mathcal{X} :\ K(x) = K^\star\}.$$

Then the expected number of competing basins compatible with the constraint satisfies

$$\mathbb{E}\Big[\, |\{\mu \neq \mu^\star :\ \mathcal{A}_\mu \cap \mathcal{C}(K^\star) \neq \emptyset\}| \,\Big] \approx \frac{M-1}{|\mathrm{range}(K)|},$$

where $|\mathrm{range}(K)|$ is the effective cardinality at the chosen discretization precision.

Remark 3 (Scaling). To drive the expected number of compatible competitors to $O(1)$, one requires $|\mathrm{range}(K)| \gtrsim M$, i.e.

$$\log_2 |\mathrm{range}(K)| \gtrsim \log_2 M.$$

This is the continuous-sector analogue of $b \approx \log_2 M$ in Hopfield clamping.

6 Reframing the Black Hole Information Problem

We now explain why the black hole information problem can be cast in the same compression–recovery shape.

6.1 Shared mathematical structure

Consider a system with:

1. microstates that scramble under chaotic dynamics into configurations that appear locally thermal;
2. global constraints or sector structure that partitions the state space into classes;
3. a decoder that can exploit global correlations and constraints to identify the correct class.

The *paradox form* arises when one assumes (incorrectly) that local thermality implies global information loss. In the compression–recovery view, local thermality is compatible with global recoverability provided sector/basin information is accessible through global correlations or boundary data.

6.2 Interpretation as sector recovery

Let M denote the number of relevant sectors/basins consistent with macroscopic observables. The theorem predicts that the information required to select the correct sector is $O(\log M)$ bits of side information, not a microstate-length description. In black hole language, the role of side information may be played by constraints encoded in correlations of the full radiation state, boundary data, or gravitational memory-like records; the decoder corresponds to whatever physical map realizes unitary recovery.

Remark 4 (What we do and do not claim). We do not claim a specific resolution mechanism (e.g. a particular holographic dictionary or an explicit physical decoder). We claim a *structural statement*: if unitarity holds and the state space decomposes into recoverable sectors, then recovery is a basin/sector selection problem whose information requirement scales with $\log M$.

7 Predictions and Failure Modes

7.1 Scaling predictions

In toy models where M can be controlled, reconstruction success should transition sharply when

$$I(K) \approx \log_2 M.$$

Below this threshold, ambiguity persists; above it, competitor basins are excluded exponentially in $I(K)$.

7.2 Failure modes

- **Degeneracy:** many basins share the same label ($g \gg 1$), reducing discriminative power.
- **Sector drift/noise:** side information is corrupted or insufficiently stable (effective $|\mathrm{range}(K)|$ shrinks).
- **Absent decoder:** dynamics do not admit a practical recovery map even if sector information exists.

8 Discussion

The contribution of this paper is to isolate a minimal, rigorous mechanism by which "scrambling" can coexist with "recoverability": recovery need not specify microstates; it suffices to specify *which basin/sector* the system should flow into, provided such basins exist and can be selected by coarse constraints.

This unifies, at the level of mathematical structure, several problems typically discussed separately: attractor recovery in associative memory, teleportation-like state transfer, and the information recovery question in black hole evaporation.

A Simulation Details

B Proof Notes and Variants

References